

LCA 4.3 – Pedigree Analysis

ENGEN370

Mark Lay

Pedigree analysis

- Assessing your LCA data quality in terms of:
 - Reliability (actual, estimated, calculated, opinion)
 - Completeness (complete – all sites, some sites, one site, unknown)
 - Temporal correlation (current, old, really old)
 - Geographical correlation (actual, larger or smaller area, different country)
 - Technology (data from that under study, related, different scale, lab)
 - Sample size (bigger is better)
- Score on a scale of 1-5 (1 being excellent, 5 being poor)
- Based on the score and category, it is given an uncertainty factor U
- You can decide what categories to use and what uncertainty factors to use (or just use the pedigree matrix factors)

Pedigree matrix



Score:	1	2	3	4	5
U1 Reliability	Verified data based on measurements	Verified data partly based on assumptions OR non-verified data based on measurements	Non-verified data partly based on qualified estimates	Qualified estimate (e.g. by industrial expert); data derived from theoretical information (stoichiometry, enthalpy, etc.)	Non-qualified estimate
	1.00	1.05	1.10	1.20	1.50
U2 Completeness	Representative data from all sites relevant for the market considered over an adequate period to even out normal fluctuations	Representative data from >50% of the sites relevant for the market considered over an adequate period to even out normal fluctuations	Representative data from only some sites (<<50%) relevant for the market considered OR >50% of sites but from shorter periods	Representative data from only one site relevant for the market considered OR some sites but from shorter periods	Representativeness unknown or data from a small number of sites AND from shorter periods
	1.00	1.02	1.05	1.10	1.20
U3 Temporal correlation	Less than 3 years of difference to our reference year (2000)	Less than 6 years of difference to our reference year (2000)	Less than 10 years of difference to our reference year (2000)	Less than 15 years of difference to our reference year (2000)	Age of data unknown or more than 15 years of difference to our reference year (2000)
	1.00	1.03	1.10	1.20	1.50
U4 Geographical correlation	Data from area under study	Average data from larger area in which the area under study is included	Data from smaller area than area under study, or from similar area		Data from unknown OR distinctly different area (north america instead of middle east, OECD-Europe instead of Russia)
	1.00	1.01	1.02		1.10
U5 Further technological	Data from enterprises, processes and materials under study (i.e. identical technology)		Data on related processes or materials but same technology, OR Data from processes and materials under study but from different technology	Data on related processes or materials but different technology, OR data on laboratory scale processes and same technology	Data on related processes or materials but on laboratory scale of different technology
	1.00		1.20	1.50	2.00
U6 Sample size	>100, continuous measurement, balance of purchased products	>20	> 10, aggregated figure in env. report	>=3	unknown
	1.00	1.02	1.05	1.10	1.20

(Ref: Energy Efficiency and Conservation Authority, Life Cycle Assessment of Electric Vehicles Final Report, 2015)

Uncertainty factors

		Uncertainty factor				
	Score	1	2	3	4	5
Reliability		1	1.05	1.1	1.2	1.5
Completeness		1	1.02	1.05	1.1	1.2
Temporal		1	1.03	1.1	1.2	1.5
Geographical		1	1.01	1.02	1.05	1.1
Technological		1	1.1	1.2	1.5	2
Sample size		1	1.02	1.05	1.1	1.2

Pedigree analysis

Data or assumption	Conventional	Electric	Qualitative description (rationale)
New Zealand electricity			For all descriptions relating to electricity below, refer also to Appendix E1.
Coal	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Data from literature-source specific to NZ electricity, described in Section 3.5 and 4.3.
Water	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Black coal	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Assumes density of coal in NZ, based on AusLCI data
Transport (trucking)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Transport (rail)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Emissions to air (carbon dioxide, methane, dinitrogen monoxide, nitrogen oxides, carbon monoxide, NMVOC, sulfur dioxide)	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Calculated, data from literature-source, based on total emissions from electricity generation using coal (2013 MftE data)
Emissions to air (other Australian NPI pollutants)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimated, based on Australian NPI data for electricity from coal
Gas	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Data from literature-source specific to NZ electricity, described in Section 3.5 and 4.3.
Water	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Natural gas	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Assumes carbon content of natural gas in NZ, based on AusLCI data
Transport (trucking)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Transport (rail)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Emissions to air (carbon dioxide, methane, dinitrogen monoxide, nitrogen oxides, carbon monoxide, NMVOC, sulfur dioxide)	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Calculated, data from literature-source, based on total emissions from electricity generation using coal (2013 MftE data)
Emissions to air (Remaining Australian NPI pollutant data)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimated, based on Australian NPI data for electricity from coal
Hydropower	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Literature source
Geothermal	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Literature source

(Ref: Energy Efficiency and Conservation Authority, Life Cycle Assessment of Electric Vehicles Final Report, 2015)

Calculating uncertainty

For each data you have assessed, use this formula to calculate an overall uncertainty (SDg95)

$$SD_{g95} = \sigma_g^2 \\ = \exp \sqrt{(\ln(U_1))^2 + (\ln(U_2))^2 + (\ln(U_3))^2 + (\ln(U_4))^2 + (\ln(U_5))^2 + (\ln(U_6))^2 + (\ln(U_b))^2}$$

The factors U_1 till U_6 referring to the scores for:

- Reliability (U_1)
- Completeness (U_2)
- Temporal Correlation (U_3)
- Geographical Correlation (U_4)
- Further Technological (U_5)
- Sample Size (U_6)

The factor U_b refers to the basic uncertainty factor and is emission-specific.

(Ref: Energy Efficiency and Conservation Authority, Life Cycle Assessment of Electric Vehicles Final Report, 2015)

Calculating uncertainty

								Pedigree ranking (1 = excellent, 5 = poor)					Uncertainty Score (SDg95)
Materials	Concrete composition (kg in mixture)			kg CO2 emissions/tonne concrete)		% difference from base case		Reliability	Completeness	Temporal	Geographical	Technological	
	Base case	Low	High	Low	High	Low	High						
Cement	126	113.4	138.6	176.0	209.1	-8.70	8.48	1	1	3	5	1	1.85
Macadam	321	288.9	353.1	199.0	186.9	3.24	-3.04	1	1	3	5	1	1.85
Gravel	469	422.1	515.9	202.1	184.4	4.80	-4.38	1	1	3	5	1	1.85
Plasticiser	0.65	0.585	0.715					1	1	3	5	1	1.85
Water	86.7	78.03	95.37	194.5	191.2	0.87	-0.86	1	1	3	5	1	1.85

When assigning rankings, make sure you give some comment as to why you assigned the ranking you did

Comparing between uncertainty and sensitivity

	Pedigree ranking	Uncertainty score (SDg95)	Sensitivity (%)	Overall
Concrete composition				
Cement	1,1,3,5,1	1.85	8.70	16.1
Macadam	1,1,3,5,1	1.85	3.24	6.0
Gravel	1,1,3,5,1	1.85	4.80	8.9
Plasticiser	1,1,3,5,1	1.85	0.00	0.0
Water	1,1,3,5,1	1.85	0.87	1.6

Used the concat function in Excel to compile the rankings into one column

Multiply uncertainty by sensitivity to get an overall value

Used conditional formatting to colour the highest values of uncertainty, sensitivity and overall values red, and lowest values green, then you can see which parameters need the most attention (i.e. most sensitivity and most uncertainty)

Overall analysis

	Pedigree ranking	score (SDg95)	Sensitivity (%)	Overall
Material impact (kg CO2/tonne)				
Limestone	4,1,1,1,2	2.11	0.02	0.0
Fireclay	4,1,1,1,2	2.11	0.01	0.0
Gypsum	4,1,1,2,2	2.13	0.00	0.0
Aggregate and gravel	4,1,1,1,2	2.11	0.11	0.2
Water	2,1,1,1,2	1.71	0.00	0.0

Overall analysis

	Pedigree ranking	Uncertainty score	Sensitivity (%)	Overall	
Limestone reaction and properties					
Calcium oxide produced per tonne limestone	4,1,1,1,1	1.83	10.07	18.4	Stoichiometric calculation
Calcium carbonate content of limestone	1,1,5,1,1	2.46	9.75	24.0	Old report
Operating parameters					
Starting dryer temperature (oC)	2,1,2,1,1	1.49	0.04	0.1	
Final drying temperature (oC)	2,1,2,1,1	1.49	0.99	1.5	
Efficiency of drier	2,1,2,1,1	1.49	1.29	1.9	
Coal CV (MJ/kg)	2,1,1,1,1	1.37	6.79	9.3	
CO2 emissions burning coal (kg CO2/kg coal)	2,1,1,1,1	1.37	6.08	8.3	
Calcliner heat required (MJ heat/kg feed)	2,1,1,1,1	1.37	4.95	6.8	

Conclusions from pedigree analysis

- Concrete composition the most important to get right, some uncertainty because data was drawn from a Masters thesis, and high sensitivity
- Calcium carbonate content of limestone had a high degree uncertainty because the data was drawn from a report in the 1970's
- Transportation had the highest certainty and lowest sensitivity
- Impact of the raw materials also had a low overall value because although the uncertainty was high, the sensitivity was very low
- Also important to get right was the Coal CV, CO₂ emissions from coal, and calciner heat required, because these had some uncertainty, but a high sensitivity

Take home messages

- Pedigree analysis is a great way of critiquing your data sources, allowing you to rank data according to an uncertainty score
- Combining the uncertainty score with sensitivity enables you to focus on which areas of your LCA need the most improvement/refinement to reduce the uncertainty
- It also enables you to identify areas you can neglect because it has a low sensitivity